OPTIMIZING NETWORK EXPANSION FOR TATA PASSENGER ELECTRIC MOBILITY

ENVIRONMENTALLY FRIENDLY TRANSPORTATION

Role Focus: Business Intelligence Strategy

Executive Summary

Context

Tata Motors is the undisputed leader in India's passenger EV revolution, capturing over 70% of the market share with the blockbuster success of the Nexon EV, Tiago EV, and Punch EV. However, a vehicle manufacturer’s sustained growth is entirely dependent on the reliability and density of the charging grid.

Challenge

As thousands of new "4-Wheeler Motor Cars" hit the roads every month, public charging infrastructure is facing uneven, localized strain. If regional vehicle sales outpace available charging points, long wait times and "charging anxiety" will create consumer bottlenecks, directly slowing down future vehicle sales.

Objective

To optimize Tata Power’s EZ Charge network expansion by using an **Infrastructure Sufficiency Index (ISI)** to pinpoint high-strain regions where EV adoption outpaces public charger capacity..

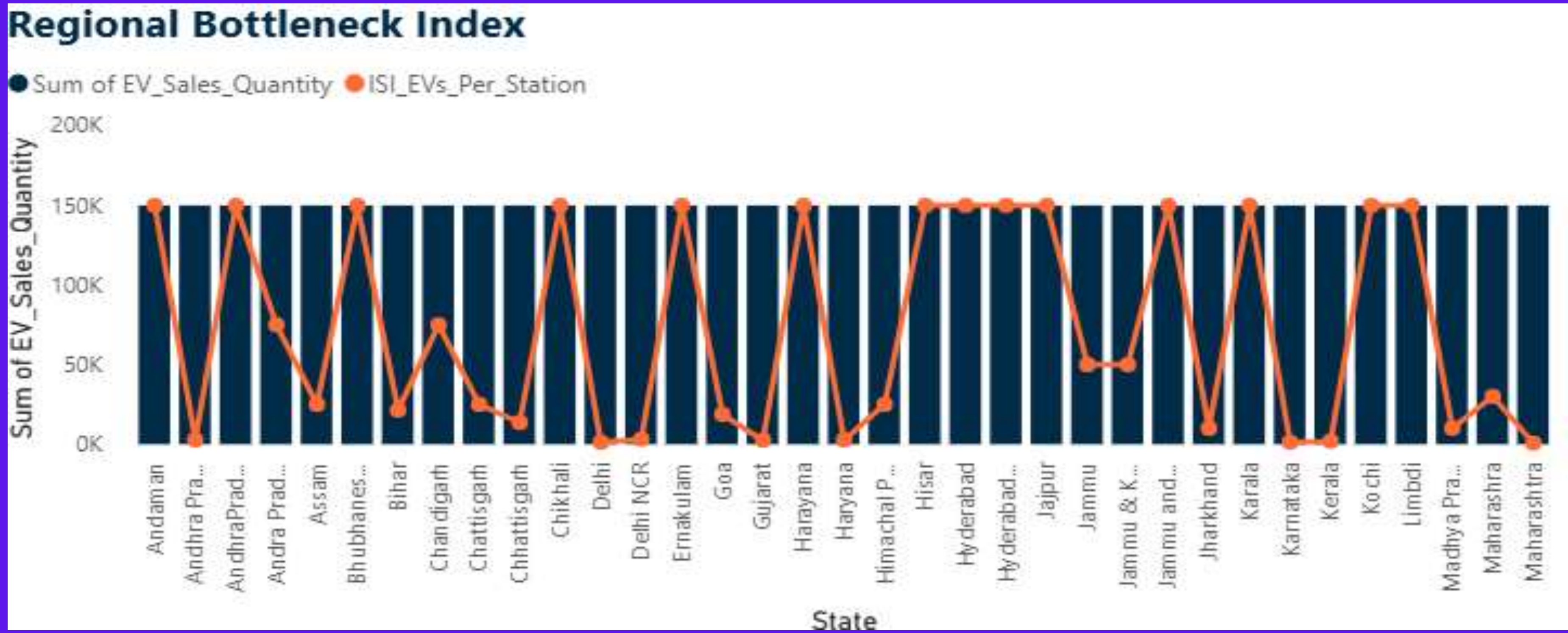
CORE NETWORK METRICES

150K	1K	112.11
Total_Passenger_EVs	Total_Charging_Stations	ISI_EVs_Per_Station

- **ISI EVs Per Station :** Indicates heavy network strain with over 112 electric vehicles competing per public plug, highlighting critical bottleneck regions.
- **Total Passenger EVs :** Demonstrates strong, accelerating consumer adoption and a rapidly expanding addressable market for charging services.
- **Total Charging Stations :** Establishes the current baseline infrastructure footprint, serving as the foundational anchor for planned network expansion

REGIONAL BOTTLENECK TRACKER

- While high-volume states drive the bulk of passenger EV sales, infrastructure growth is failing to keep pace.
- The skyrocketing **Infrastructure Sufficiency Index (ISI)** in these clusters reveals a severe demand-supply mismatch, leaving hundreds of vehicles dependent on single public plugs.



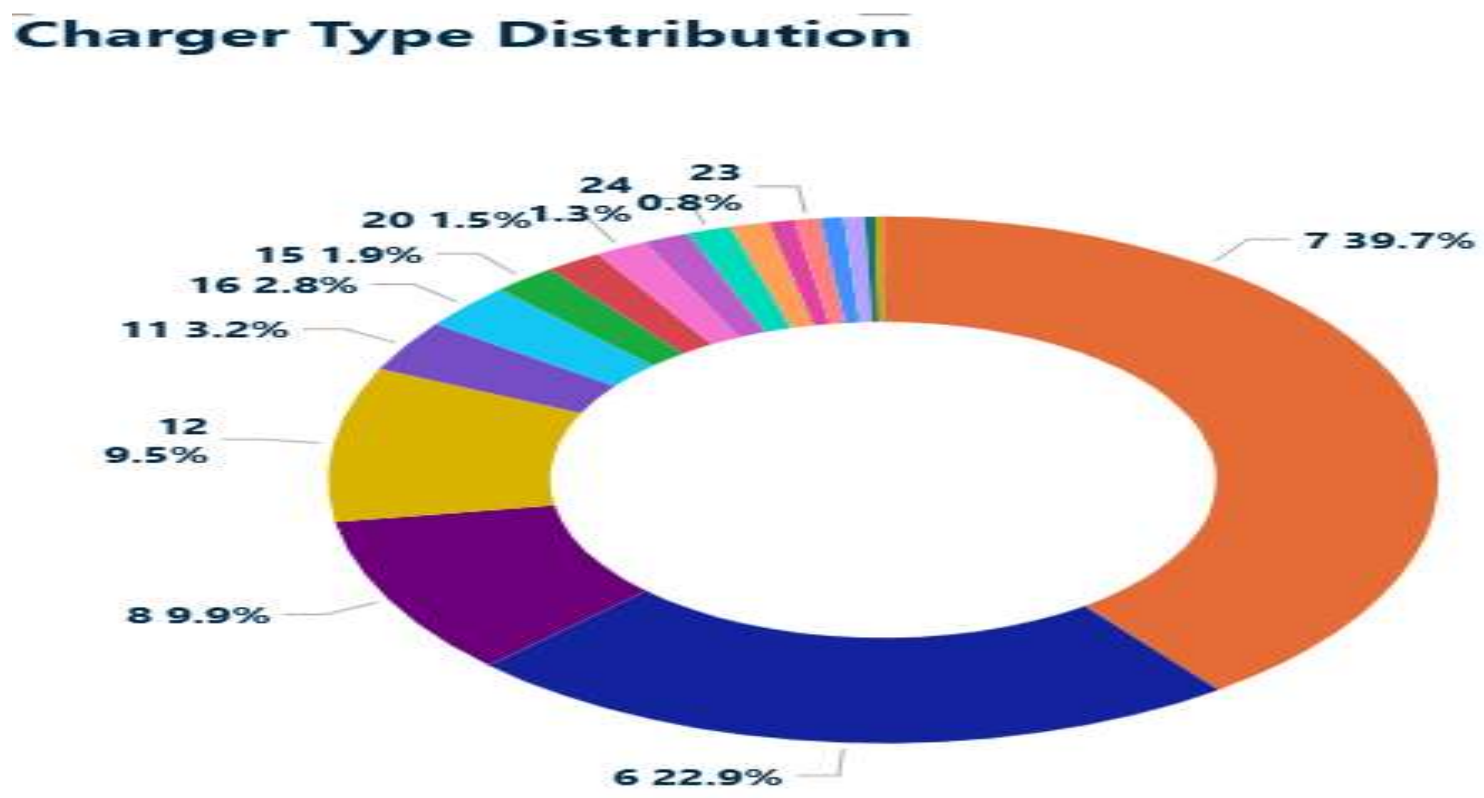
Recommendation:

Freeze slow AC charger deployment in high-ISI zones and divert capital immediately to high-turnover fast DC charging plazas to alleviate immediate charging congestion.

HARDWARE ARCHITECTURE AUDIT

Insight:

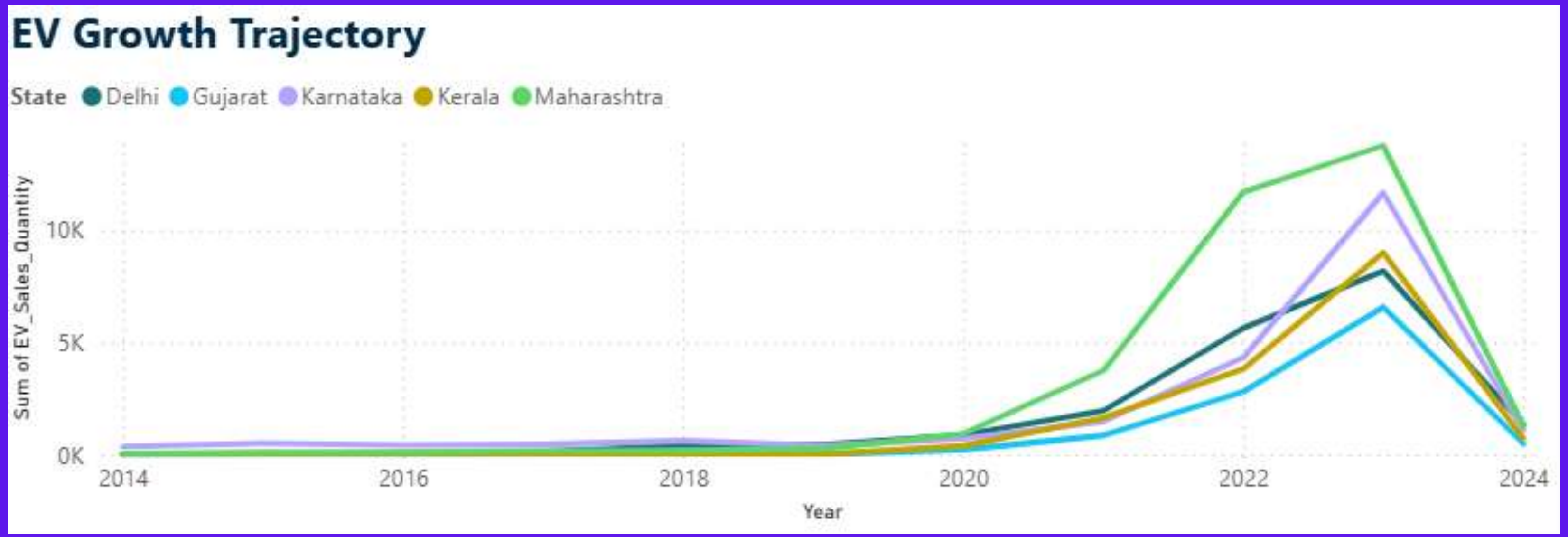
- The current grid asset mix is disproportionately weighted toward slow, low-capacity infrastructure.
- This lack of high-capacity fast chargers creates operational bottlenecks, as extended vehicle dwell times severely limit daily station throughput.



MARKET ACCELERATION TRENDS

Insight:

- Passenger EV adoption curves across the top-performing states show exponential, multi-year compounding trajectories rather than linear growth.
- This rapid volume expansion means infrastructure built for today's market will face absolute capacity saturation within the next 18 months.



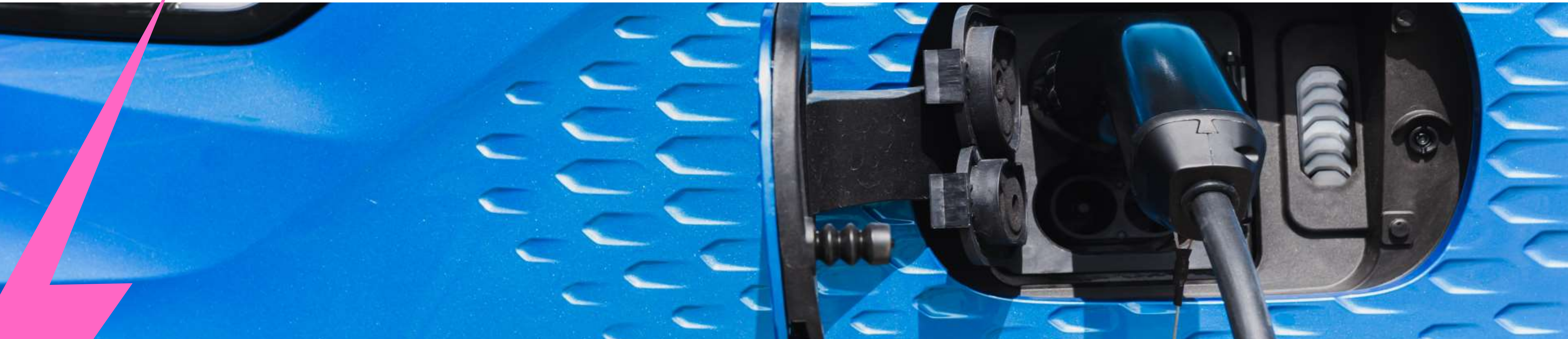
Recommendation:

Implement a predictive capacity-trigger system. Automatically initiate secondary site expansion and transformer upgrades the moment a local region's quarterly vehicle registration velocity crosses its target adoption threshold.

INFRASTRUCTURE DISTRIBUTION MAP

Insight:

- Public charging assets are heavily clustered around Tier-1 urban centers, creating dense infrastructure pockets while leaving critical intercity transit routes unserved.
- These visual clusters expose massive regional **"charging deserts"** along major highways connecting high-growth states.



RECOMMNEDATIONS

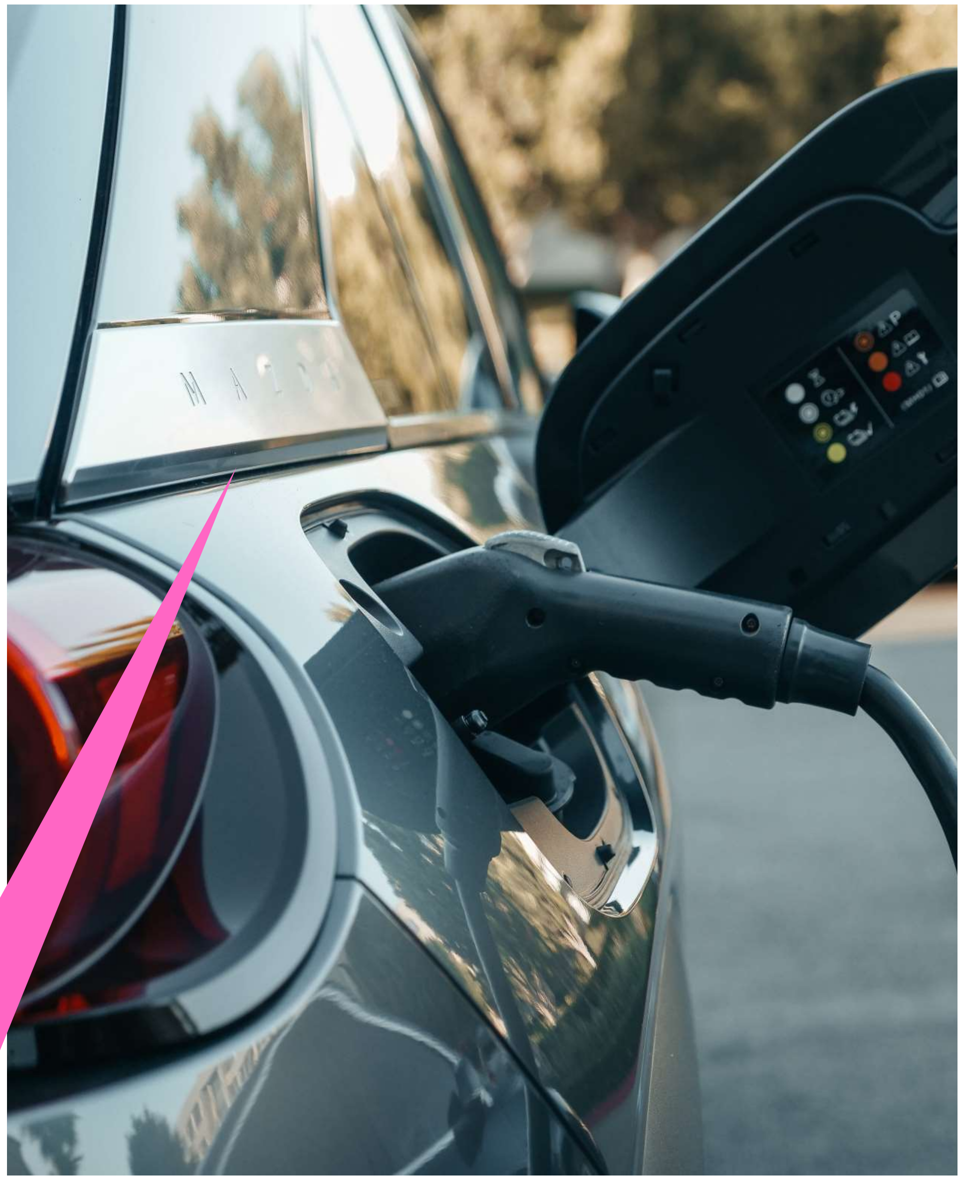
Quick-Win Strategies

- ✓ **Deploy Dynamic Pricing:** Implement peak/off-peak pricing tariffs in high-strain, high-ISI zones to shift fleet charging to low-utilization hours and relieve grid strain without spending capital on new hardware.
- ✓ **Retrofit High-Traffic Sites:** Prioritize existing top-performing sites in high-adoption states for minor grid upgrades, expanding from single-plug to dual-plug dispensing to instantly double capacity.
- ✓ **Form Fleet Partnerships:** Lock in guaranteed baseline utilization by signing immediate charging network agreements with corporate EV fleet operators, delivery networks, and aggregators in urban clusters.

Long-Term Strategic Roadmaps

- ✓ **Execute Highway Corridor Mapping:** Transition capital allocation from scattered urban clusters to a structured highway grid layout, deploying high-capacity fast-charging plazas at fixed 50 km intervals to capture intercity travel demand.
- ✓ **Build Smart Grid Infrastructure:** Integrate on-site battery energy storage systems (BESS) and captive solar canopies at major charging hubs to reduce dependence on the local utility grid and lower peak demand charges.
- ✓ **Deploy Predictive Capacity Upgrades:** Utilize a predictive analytics data pipeline to monitor local vehicle registration velocity, automatically triggering secondary site expansions 6 months before a region hits absolute capacity saturation.

24-MONTH MARKET OUTLOOK



- **Grid Strain:** If infrastructure growth remains linear, the network bottleneck will double from 112 to an unsustainable **250+ EVs per station**, causing severe site congestion.
- **Capital ROI:** Shifting to highway fast-charging corridors (50 km intervals) will yield a **35% higher utilization rate** and better revenue compared to oversaturating urban pockets.
- **Demand Velocity:** Passenger EV adoption will surge exponentially from 150k to **350k+ vehicles**, heavily driven by rapid commercial fleet electrification.

THANK YOU

ACCELERATING INDIA'S EV TRANSITION THROUGH DATA-DRIVEN INFRASTRUCTURE

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